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PROJECT REPORT 2022:

**TOPIC: REMOTE AUTOMATIC GATE
CONTROLLER FOR HOMES**

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Abstract

This project aims to design and construct a remote automatic gate controller for homes using Bluetooth communication, Global System of Mobile Communication network and a manual override system. The need for efficient control of access for persons, especially physically challenged and the aged and vehicles going in and out of the home has placed a high demand for automatic gate. In this work, the authors took short range and long range communications when overhauling for the remote-control aspect of the project objective. They presented a developed GSM and Bluetooth operated main controller for automatic operation of sliding gate. Circuit configuration and component description of the system were presented. Finally, the system was tested to ensure it works properly and meets specifications.

Keywords: Automatic gate, Remote Control, Bluetooth, GSM, sliding gate, Manual Override.

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1.INTRODUCTION

Background to study

The security challenges encountered in many places require efficient means of controlling access to high input areas such as communities, recreational areas, offices and homes preventing unwanted access[1]. The prevalence of armed robbers and other groups that constitute a nuisance to the society has led to the loss of lives and properties worth millions of money. Pointing that many attacks that led to significant loss of lives and properties have occurred in unguarded homes [2] Postulated that since the development of early societies and housing complexes humans have attempted to regulate their own built environment[3].

A gate is a point of entry to a space enclosed by walls or an opening in a fence. Gates may prevent or control entry or exit or they may be merely decorative[4]. Gates play a vital role in providing controlled access to specific areas of interest such as homes. They are mainly security oriented. First gate introduced in the world was completely manual and humans were required to open such gates[5]. It required physical work.

Study Problem

In modern gates, complexities are introduced as a measure to increase security. These complexities include increased thickness of material, weight, size and complex locking mechanisms. The additional complexities cause increase in access time and compromises accessibility in manual-operate gates: A homeowner coming back home from work or during an emergency will have to use time to get in and out of his car to open and close the gate or wait for an operator to open it. It makes the gate operation time-consuming. Also, manual gate uses a sizeable human effort in opening and closing of the gate. The use of human effort in opening gates requires human resources. People in wheelchairs, aged and other disabled individuals in homes that access the gate frequently, do not have that capability required to open these gates manually. Boyce and Shields et al. (1999)[6] presented the results of an experimental program to determine the ability of disable people to negotiate doors with closing force that range from 20N to 70N. Their results indicate many disabled people find it impossible to manual gates even at the relatively low forces. Due to these factors, there has been advancement in gate creation known as automatic gate.

Justifications

Automation is the art of making processes or machines self-acting or self-moving; automatic[7]. Automation in the electrical, electronics and computing world has grown rapidly of which dates back to 1940 when the first electronics computing machine was developed[8]. This has aided humans as it basically reduces/eliminates human intervention of which automatic gate also makes the list of electro-computing world.

Automatic gate consist of two basic components: Gate opener and Gate operator/controller. The gate is the physical object that is moved to block the gate opening while the gate controller is the

machinery that moves the gate in and out of the gate opening. Automatic gates can be operated by a variety of access control devices including portable wireless transmitter (i.e. remote control unit), card readers, vehicle tag readers and digital keypads[9]. The theory of an automatic gate is simple. When the remote control unit is pressed a signal is sent to the gate operator to open and close the gate.

Installing an automatic gate helps prevent people from gaining access at will. Thus a higher level of security is ensured. Remote control unit work to open and close automatic gates from a distance. This eliminates the access/waiting time encountered during manual operation, ensures accessibility to mainly disabled or aged individuals in the home.

Gates are commonly now used in residential areas. For convenience and security, automatic gates have been on the increase in homes.

Aim:

The aim of the project is to design, construct and test a remote automatic gate controller for homes.

Objectives:

- Select a standard sliding gate
- Select and size a motor
- Select a wireless communication technology
- Design, construct and test a controller for the motor and remote control unit based on the wireless communication technology for automatic opening and closing of the gate.

2.LITERATURE REVIEW

A significant number of literatures on remote automatic gate controller technologies have been reviewed. The rationale for the reviews is to enable us to be abreast with recent works on controlling gates automatically with the help of remote control units. It serves as guide for our project.

Relevant Papers related to our project:

2.1. Design of Automatic Gate Control Using Infrared Remote with Password Protected Features by Ayodele Sunday Oluwode et al [10] in 2014.

In this paper, the authors designed an automatic gate using an infrared remote with a password protection feature. An eight-bit microcontroller, AT89C2051 was used as the main controller to the motor. The microcontroller received an infrared pulse from LED transmitter of a remote control handset through an IR Sensor. It decoded the pulse and switched ON a relay that controlled the sliding gate DC motor to either open or close the gate. When an incorrect password was entered, the gate did not open and an integrated alarm system sounded.

Merits

1. The system is cheaper to implement.

Demerits

1. Infrared communication used as the wireless remote control mechanism is line-of-sight independent and too short ranged.

2.2. Implementation of Intelligent Automated Gate System with QR code by E. Hamid et al [11] in January 2018.

In this paper, the authors designed and implemented a medium level security gate device for small agencies that cannot find the money to put in high-tech auto gate systems. The main components included a PIR motion sensor, servo motor, Arduino Uno Microcontroller, Piezo buzzer, an internet digital camera related to the PC thru USB cable and a software thing carried out using VB.NET. An interface was created using VB.NET to control the Arduino Uno Microcontroller. If an authorised QR code scan is detected via the internet web camera, a rotation command is sent from the VB.NET software interface to the Arduino Uno microcontroller through a serial port. The motor is rotated at 90° and the gate is opened. If an unauthorized scan is detected, the piezo buzzer is triggered and electronic mail is sent to the security department. It can detect the movement of passing vehicles. Once the movement is detected, the PIR sensor actuates the gate motor to shut the gate. All data of the group of workers was saved into SQL database wherein workers' codes are created and their incoming and outgoing statistics recorded.

Merits

1. The QR code recognition system is 99% accurate.
2. The QR code developed is not easily cracked and the system security level is increased.

Demerits

1. The VB.NET software used is slow in application processing. Thus, it will make the entire system slow.

2.3. Smart Gate by Prof. Hemlata Ohal et al [5] in 2018.

In this paper, the authors proposed a system which included a combination of sensor technology, and IOT (Internet of Things). Using this system one could control the gate through his smartphone. The primal working consisted of front end components i.e. the RFID Tags and reader. The RFID tags had unique ID that detected an authorized user. If the user was unauthorized, a buzzer ringed to alert the security. When the unauthorized user was a guest, a mobile application was be used. The mobile application was connected to the Arduino Microcontroller with the help of the Bluetooth Module HC-05. The mobile application worked as a manual alternative to the automatic system. The system also worked for vehicles. When a user entered with his car, the system checked the vehicle number and displayed if the user was authorized or unauthorized and request access. If access was not provided, the guard can granted access with the help of buttons on the mobile application. There are two buttons on the mobile application; one for the incoming vehicles and the other for outgoing vehicles. This system used Arduino IDE Programming and Embedded C Programming for uploading the codes. It also used the android studio for developing android/mobile applications required to operate the system manually.

Merits

1. System is not easily hacked.

Demerits

1. In the gate control unit, the RFID tag option is not remote controlled and unreliable.

2.4. Design and Implementation of an Automated Security Gate System using Global System for Mobile Communication Network by Promise Elechi et al [12] in 2021.

In this paper, an automated security gate using the Global System of Mobile Communication Network was designed and constructed.

The gate was operated using SMS communication between the user, the control panel to be installed in the house and the gate drive system installed at the gate. When the automatic gate control system was powered on, the service provider of the SIM-card network connected to the gate system by the SIM800L GSM modem and the system was ready to receive commands from the user. The code instructions were read and executed by the ATmega328 Microcontroller. The microcontroller gives the command to a ULN2003 stepper motor which opened or closed the gate. The control panel had a manual push-button switch to reset the system. LEDs were added

to the system to indicate gate operation. It lighted Yellow when the gate begun to open and Green when the gate was about to close. The system hardware materials included SIM800L GSM Modem, ATmega328 Microcontroller, Power Supply 230/12 1.5A, ULN2003 Stepper Motor Driver, Motor 12V, 200mA, Atmel AVR, Mechanical Fabrication tools, Wiring cables/Accessories and the software materials include Fritzing software, Proteus 8.9 software, Arduino Software IDE, C Programming software.

Merits

1. Automated gate controllers using mobile phone GSM offers a unique, secure and convenient way of controlling access to facilities.

Demerits

1. The system is not beneficial for short range communication.

2.5. An electronically controlled automatic security access gate by Jonathan A. Enokela et al [2] in November 2014.

In this paper, the authors designed an electronically controlled automated security access gate proposed in this work prevented unwanted access to controlled environment. This was achieved mainly through the use of a Radio Frequency (RF) transmitter-receiver pair. In the design, a microcontroller was programmed to decode a given sequence of keys that were entered on a keypad and commanded a transmitter module to send out this code as the signal at a given radiofrequency from a distance of 7meters. Upon reception of this RF signal by the receiver module, another microcontroller is activated by a driver circuitry to operate the gate automatically. The security access gate control system consisted of two principal units: a handheld transmitting unit and receiving unit. The transmitting unit consisted of a 3x4 matrix keypad, a microcontroller and an RF transmitter module. The receiving unit consisted of an RF receiver module, a microcontroller, a motor driver and a motor. The programs for the transmitting and receiving units were written in C language and were built and debugged in the KEIL micro vision 4 integrated development environment. The simulation of the firmware and hardware was carried out using the Proteus Virtual System (VSM) modeling environment version 7. The resultant hex file was burnt into the program memory of the microcontroller with the aid of a TOP2005 universal programmer.

Merits

1. The inclusion of a password inserted makes this design have enhanced security.

Demerits

1. The distance for communication between the transmitter and receiver is too short.
2. The access gate control system is not built to hold a heavy gate

2.6. Implementation of Home Security System using GSM module and Microcontroller by Abishek S. Parab et al [13] in 2015.

In this paper, the authors mainly focused on the security of the home where the user is out of place. GSM based technology was proposed to update the owner of an intruder or danger to the house. Suspected activities were conveyed to the remote user through SMS or Call using GSM technology.

The door of the house contained a magnet which was connected to a relay. The relay and magnet combination was installed at the entry point to the GSM-based security system which was installed on the door and produced a signal through a public telecom network. If the door was closed or there was no intruder, no action was taken and the LED glowed yellow. As an intruder opens the door, the relay was disconnected /detached from the magnet, the LED glows with RED light and signals are generated via relay and sent to the 8051 microcontrollers. The microcontroller would take appropriate action according to the piece of code written (burnt) in it. Through this action, the GSM module would be activated and an SMS sent to the phone number declared in the piece of code embedded in the microcontroller.

The system contained a GSM Modem, Magnet, Relay, 8051 Microcontroller with C programming language embedded within it, LEDs and Cell-phone or device in which GSM technology can be used. Flash software was used to burn the program into the microcontroller.

Merits

1. The system responds quickly to a detected intruder and the GSM module will alert the homeowner in time.
2. The user can get alerts anywhere through the GSM technology thus making the system location independent.
3. Communication through SMS works on any mobile network.

Demerits

1. There is no mechanism to ensure that the person opening the door is not an intruder but a friend or family member.

3. THEORY & DESIGN CONSIDERATIONS

In this chapter, we discussed the working our project.

3.1. WIRELESS COMMUNICATION TECHNOLOGY

Wireless communication is the fastest growing and most vibrant technological areas in the communication field. Wireless communication is a method of transmitting information from one point to another without using any connection like wires, cables or any physical medium within a shorter distance or across the globe (i.e. long range). Generally in a communication system information is transmitted from transmitter to receiver that is placed over a limited distance. With the help of Wireless Communication, the transmitter can be placed anywhere between a few meters (like a T.V. Remote control) to a few thousand kilometers.

3.1.1. Short Range Wireless Communication Technology

Short range wireless communication systems are systems that provide wireless connectivity within a local sphere of interaction. Short range wireless communication is of the types: Bluetooth, Wi-Fi, ZigBee, UWB, Infrared. Comparing the various technologies;

	Bluetooth	ZigBee	Wi-Fi	Infrared
Data rate	1 Mbit/s	20,40 and 250 kbits/s	11 and 54Mbit/s	115kbits/s
Range	10m	10 to 100m	Up to 100m	5m
Networking Topology	Ad-hoc Small networks	Ad-hoc, peer to peer, star or mesh	Point to hub	
Frequency	2.4GHz	868 Hz (Europe) 900 to 928MHz (North America) 2.4GHz (worldwide)	2.4 and 5GHz	Zz
Power Consumptions	Low	Very low	High	

Typical applications	Inter-device wireless connectivity eg. Phones, headsets, laptops.	Industrial control and monitoring, buildings automation, toys, games	Wireless local-area network (WLAN) connectivity, broadband internet	
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Choice of Short range communication:

Our choice of short range communication is based on the power, cost and easy implementation as a handset remote control based on the comparison of the short distance technologies above. Bluetooth is chosen as the wireless communication technology for short range communication.

It is chosen mainly because:

- o It has a communication range of 10m which is good for small network communication with the gate.
- o It can be designed to transmit/receive data small amounts of data on a fairly infrequent basis, all while consuming extremely low amounts of power
- o Bluetooth products can only be powered from a coin cell battery. Thus the overall cost of product is low and may have a longer battery life.

3.1.2. Long Distance Wireless Communication Technologies

The term “long distance communications” arose in the technology of transmission of electrical signals over wire communications lines (intercity communications). These technologies in opposition to more traditional communication technologies rather defined as “short range”, allow kilometer wide wireless communication.

Choice of Long Distance Wireless Communication Technology

In this project, Mobile Phone-Based gate access using GSM communication is chosen to enable gate access to close friends and family that will need it while the homeowner is on a trip or not available.

GSM communication is chosen because:

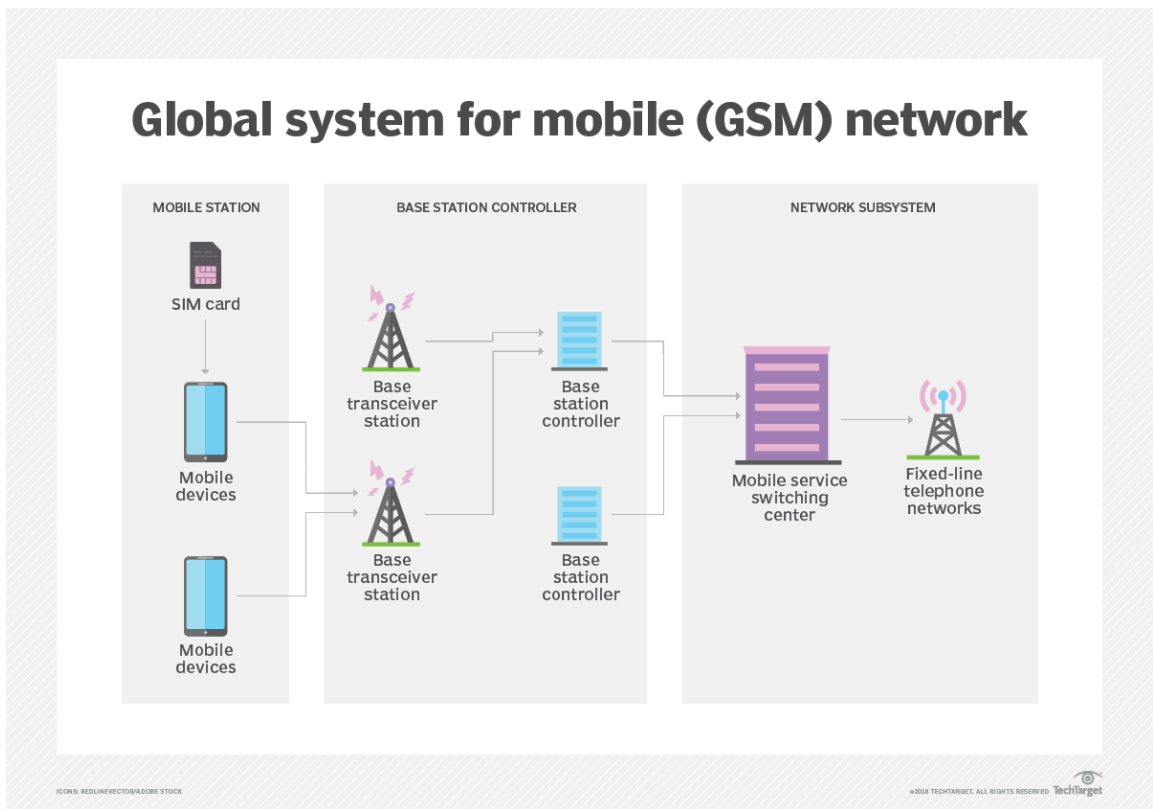
- It enables opening and closing of the gate by SMS communication.
- Gate access enabling is done with low cost mobile sets and base stations

GSM

GSM stands for Global System for Mobile Communication. It is a digital mobile network that is widely used by mobile phone users in Europe and other parts of the world.[tutor] GSM is an

open and digital cellular technology used for transmitting mobile voice and data services operate at the 850MHz, 900MHz, 1800MHz, and 1900MHz frequency bands. It is used for transmitting mobile voice and data services[14].

The GSM network has four parts that work together to function as a whole: the mobile device itself, the base station system (BSS), the network switching subsystem (NSS), and the operation and support subsystem. The mobile device connects to the network via hardware. The subscriber identity module (SIM) card provides the network with identifying information about the mobile users[15].

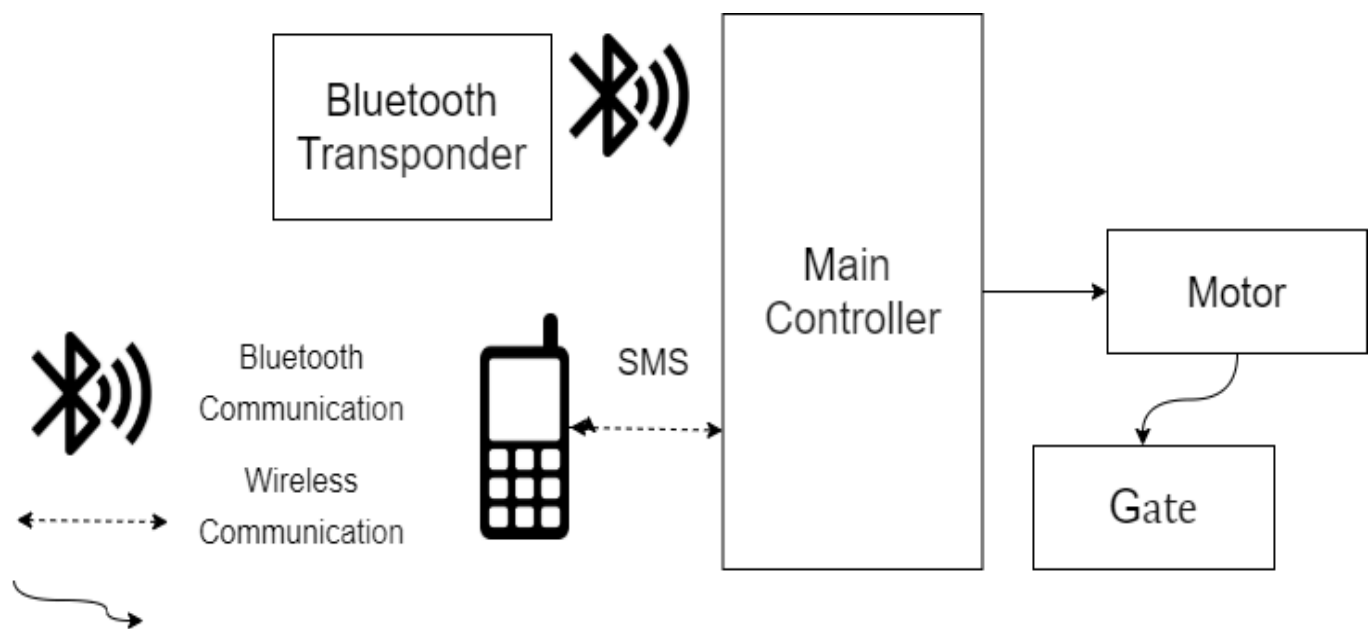


The BSS handles traffic between the cellphone and the NSS. It consists of two main components: the base transceiver station (BTS) and the base station controller (BSC). The BTS contains the equipment that communicates with the mobile phones, largely the radio transmitter receivers and antennas, while the BSC is the intelligence behind it. The BSC communicates with and controls a group of base transceiver stations.

The NSS portion of the GSM network architecture, often called the core network, tracks the location of callers to enable the delivery of cellular services. Mobile carriers own the NSS. The NSS has a variety of parts, including mobile switching center (MSC) and home location register (HLR). These components perform different functions, such as routing calls and Short Message Service (SMS) and authenticating and storing caller account information via SIM cards.

3.2. OVERVIEW OF OUR PROPOSED SYSTEM

The block diagram below shows the block diagram overview of our system and how subsystem is interconnected.



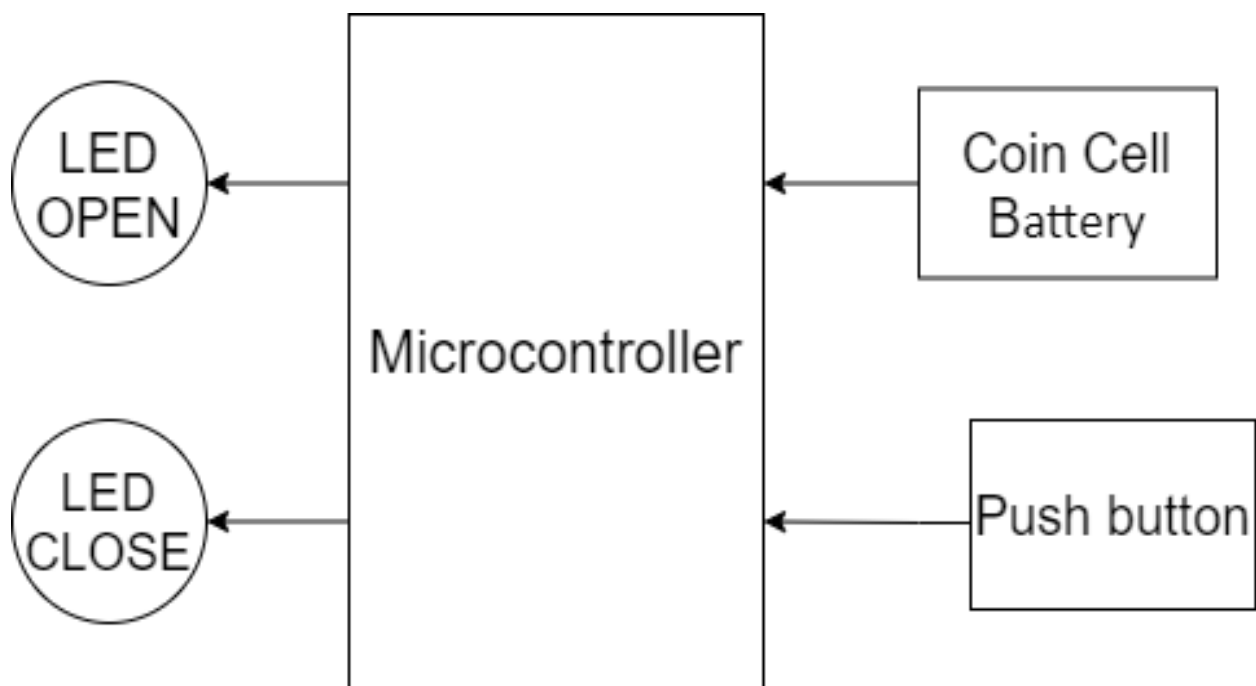
Overall Working of Proposed System:

- It is mainly made up of 4 main parts:
 - o Main controller for the motor
 - o Bluetooth transponder
 - o Mobile Phone using GSM communication
 - o Selected Motor and Gate

The foremost working consists of the front end components i.e. Bluetooth transponder and Mobile Phone for GSM communication through SMS. The Bluetooth transponder has unique UUID that is transmitted once the push button of the transponder is pressed and the main controller detects it. If the main controller authorizes the user, then control is shifted to the motor to operate the gate. Bluetooth transponder is used for short range, 10m communication by legitimate individuals in the home.

For instances where there is no one in the home and the homeowner is far-off the field of the home, SMS communication is used. SMS message is sent from the mobile phone the main controller to control the motor to open the gate.

3.2.1. BLUETOOTH TRANSPONDER



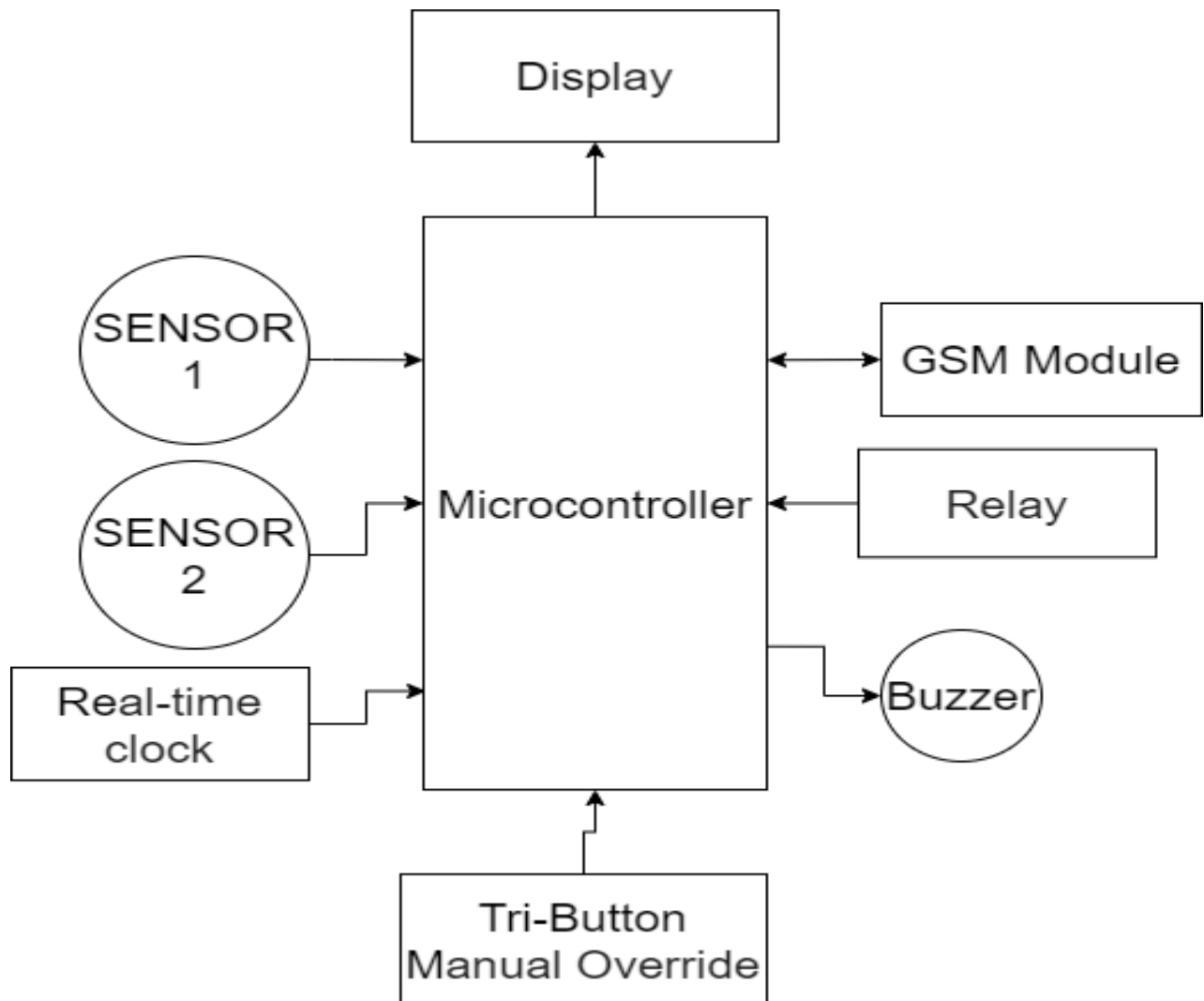
Working of the transponder

The Bluetooth communication is made possible by our custom design Bluetooth transponder. A transponder is a wireless communication monitoring or control device. It is the contractions of the words transmitter and responder.

The Bluetooth transponder is issued to only legitimate users like family and close friends. It communicates with the main controller to open the gate within 10m range to the gate. The Bluetooth transponder is simple and handheld.

Each Bluetooth transponder is encoded with a unique UUID that is transmitted to the main controller when a button on the transponder is pushed. On the reception of the signal the main controller verifies the UUID by comparing the received UUID to the stored UUIDs stored in its EEPROM. Once verified, the main controller initiates gate open activity and sends a time stamped notification to the administrator signaling entry. Since the user is most of the time not in a close contact with the gate, light-emitting diodes connected to the microcontroller of the transponder lights to indicate the gate activity.

3.2.2. Main Controller



Working of the Main Controller

The main controller is the circuitry for the motor to enable automatic gate opening. The main controller enables Bluetooth communication from the Transponder and SMS communication from the GSM modem/ Mobile Phone through the GSM Module. Upon receiving and verifying signal from either the Transponder or SMS from the Mobile Phone, current flows to the relay to power the motor and the gate is opened. Once the gate is opened, 2 sensors positioned at the front and back of the gate is triggered to sense entry.

Aside grating entry, the SMS protocol is also used to send time stamped notification to signal the administrator on the gate activity (i.e. opening and closing). Real time clock is used as the time keeper and enables the right time notification to the administrator. A display is employed as an additional feature. It is placed on the main controller to also display the gate activity (i.e. who accessed the gate and when it was accessed).

In the event of power failure, the gate will not be able open or it will get stuck if it was opened. A special feature is designed to shift control to manual work during power outage. The feature is called Tri Button manual override. It is an arrangement of 3 push button programmed in a certain pattern. When the correct pattern is pressed, control is shifted.

3.3. The Bluetooth Transponder

3.3.1. Microcontroller

The microcontroller is the main processor of the transponder. It must enable Bluetooth communication.

Options:

- o Arduino Microcontrollers
- o ESP8266 Microcontroller
- o ESP32 Microcontroller
- o PIC Microcontroller

Choice of Microcontroller

For this project, ESP32-WROOM-32D module Microcontroller is chosen as the main processor for the transponder due to its:

- o Integrated Bluetooth functions. It does not need an additional radio modules or Bluetooth modules like in Arduino and other microcontrollers.
- o Faster execution speed
- o Low price and small package
- o Larger programmable Flash memory space



Specifications/ Technical features

Microprocessor	Single or Dual-Core 32-bit Tensilica Xtensa LX6
Maximum Operating Frequency	240MHz
Operating Voltage	3.3V
Analog Input Pins	12-bit, 18 Channel
DAC Pins	8-bit, 2 Channel
Digital I/O Pins	39 (of which 34 is normal GPIO pin)
DC Current on I/O Pins	40 mA

Microprocessor	Single or Dual-Core 32-bit Tensilica Xtensa LX6
DC Current on 3.3V Pin	50 mA
Memory	520 KB of SRAM, 448 KB of ROM and 16 KB of RTC SRAM.
Communication	SPI(4), I2C(2), I2S(2), CAN, UART(3)
Wi-Fi	802.11 b/g/n
Bluetooth	V4.2 – Supports BLE and Classic Bluetooth

3.3.2. Light-emitting diode (LED)

A light-emitting diode (LED) is a semiconductor device that emits light when an electric current flows through it. When current passes through an LED, the electrons recombine with holes emitting light in the process. It is employed as indicator lamps for the gate operation.

Types of LED include:

- Dimmer switches
- Color LED
- LED Tubes for lighting
- SMD LED
- COB LED and many more.

Due to the nature of work, we need an LED to be mounted on a chip. This includes: COB LED and SMD LED.

- o COB (Chip On Board) LED

COB LEDs are basically multiple LED chips (typically nine or more) bonded directly to a substrate by the manufacturer to form a single module. A COB LED light features one mounted LED chip.

Advantages

1. Being multi-chip packaged, the light emitting area of a COB LED can contain many times more light sources in the same area that standard LEDs could occupy resulting in a greatly increased lumen output per square inch.
2. Soldering of the individual LED chips is not necessary as each chip is directly mounted onto the substrate. The smaller amount of welding points results in lowering the rate of failure.

Disadvantages

1. The large size of the heat sink to cool the chip causes the COB LED to cost a little more.
2. It has a lower color uniformity, and it can result in a poorer viewing experience

SMD (Surface Mounted) LED

SMD LED Light stands for “Surface-Mount-Device,” which is a Light-Emitting-Diode. It is a particular type of LED that is known and differentiated not only based on its physical dimensions, but also for having a 3-in-1 encapsulation .That is to say, an SMD LED integrates three colors all at once: Red (R), Green (G), and Blue (B).

Advantages

1. More bright light is produced with less consumption of power and wattage
2. It is incredibly small

Disadvantages

1. SMD LED display is more prone to damage because the LEDs are smaller.

Choice of LED

SMD LED is chosen as the LED gate activity indicator. It was chosen because of its small size as the Bluetooth transponder design is small and lower power consumption.



3.3.3. Battery

To power up the Transponder, battery is used. The type of batter applicable for Transponder is coin/button cell battery.

Coin/ button cell battery

A coin/button cell battery is a small, circular, stainless steel disc that provides power to portable devices, like watches, remote control and computers. **Coin** batteries (or cells) contain lithium chemistry and have a 3 Volt output. Button batteries are similar to coin cell batteries but are

smaller in shape. Button batteries are made from a variety of chemistries, including silver oxide, alkaline, lithium, and zinc air and have 1.5 Volt output.

Simply put, coin cells are wider and slimmer like a coin, and button cells have a narrower diameter and are taller and stockier in appearance.

The Transponder is simple and handheld so requires a small battery but with enough power.

Choice of Battery

The best choice for our design is 3 Volt 12mm lithium coin cell battery. It is selected because of its small size, low cost and low power consumption.



3.3.4. Push Button

A push button switch is a mechanical device used to control an electrical circuit in which the operator manually presses a button to actuate an internal switching mechanism. They come in a variety of shapes, sizes, and configurations, depending on the design requirements.

Types of Push Button

- Momentary
- Latching
- Electric
- Pneumatic

Choice of Push Button

In order to enable opening of the gate once the push button on the Transponder is pressed, a momentary push button switch is chosen

Momentary Push Button Switches

Momentary push button switches are non-latching switches which cause a temporary change in the state of an electrical circuit, only while the switch is physically actuated.

There are two types:

- o Normally Open (NO) switch: It is known as a push to make switch which allows electricity flow between its contacts when held in. When the button is released, the circuit is broken. It is used in doorbell.
- o Normally Closed (NC) switch: It is known as a push to break switch which breaks electricity flow in a circuit when it is pressed.

In the working our system, the gate is initiated only when the button is pressed. Therefore, a Normally Open push button is chosen.



3.4. Main Controller

The main controller is the main circuitry that drives the motor and all other functions including GSM communication.

3.4.1. Sensor

It is a type of device that detects and responds to some type of input from the physical environment. The specific input could be light, heat, motion, moisture, pressure or any one of a great number of other environmental phenomena.

The types of sensors for automatic gate

- Radar Movement Sensors
- Pressure Sensors
- Optical Sensors or Motion Detectors
- Passive Infrared Movement (PIR) Sensors
- Active Infrared Sensors

Choice of Sensor

In our design, we chose passive infrared motion sensors to close the gate after the user has passed completely.

- Active Infrared motion sensors

Active IR sensors have two parts: a light emitting diode (LED) and a receiver. When an object comes close to the sensor, the infrared light from the LED reflects off of the object and is detected by the receiver. Active IR sensors act as proximity sensors, and they are commonly used in obstacle detection systems (such as in robots).



3.4.2. Microcontroller

Microcontroller is a compressed microcomputer manufactured to control the functions of embedded systems in office, machines, robots, home appliances, motor vehicles and a number of gadgets. Microcontroller is comprises components like - memory, peripherals and most importantly a processor. Microcontrollers are basically employed in devices that need a degree of control to be applied by the user of the device.

Options

- 8051 Microcontroller
- PIC Microcontroller
- Arduino Microcontroller
- ESP Microcontroller

Choice of Microcontroller

In this design, ESP Microcontroller is chosen as the main processor of the system to control the motor. ESP boards are lower-cost and higher performance than competing microcontrollers. There are mainly two types of ESP Microcontroller: ESP8266 and ESP32.

Comparison Table between ESP32 and ESP8266

Comparison	ESP32	ESP8266
Clock Frequency	160 or 240mhz	80 mhz
Bluetooth	BLE	No

Hall Sensor	Yes	No
Temperature Sensor	Yes	No
Touch Sensor	10	No
Low Power Consumption	10uA deep sensor	20uA
Total GPIO	34	17
SPI	4	2
Microcontroller	Single or dual-core 32-bit LX6 Xtensa	Single core 32-bit L106 Xtensa
ROM	448 KB	No
FLASH	520KB	32KB

Upon comparison of the two types, ESP32 is chosen as the main processor in the main controller for the motor. It is chosen mainly because of its integrated Bluetooth function. Bluetooth communication between the transponder and main controller for the motor can only be established if the main processor supports it. It also provides faster processor and good memory size, which allow considerable larger projects to be designed on only one SOC.

ESP32 WROOM Module Microcontroller

ESP32 is a low-cost, low-power Microcontroller with an integrated Wi-Fi and Bluetooth. Espressif Systems ESP32-WROOM-32 MCU Modules are powerful and generic Wi-Fi/BT/BLE MCU modules that target a wide variety of applications. These modules target applications ranging from low-power sensor networks

SPECIFICATIONS

- 802.11 b/g/n (802.11n up to 150Mbps) Wi-Fi protocol
- Bluetooth v4.2 BR/EDR and BLE
- SD card, UART, SPI, SDIO, I²C, LED PWM, Motor PWM, I²S, and IR module interfaces
- Hall sensor
- 2.7V to 3.6V operating voltage supply
- 80mA average operating current
- -40°C to 85°C operating temperature range

3.4.3. GSM Module

A GSM module is a chip or circuit that will be used to establish communication between a mobile device or a computing machine and a GSM or GPRS system. GSM module can perform many operations like receive, send or delete the SMS on SIM, make and receive or reject the voice call. There are various modules that have been implemented for GSM communication.

Some of the considered are

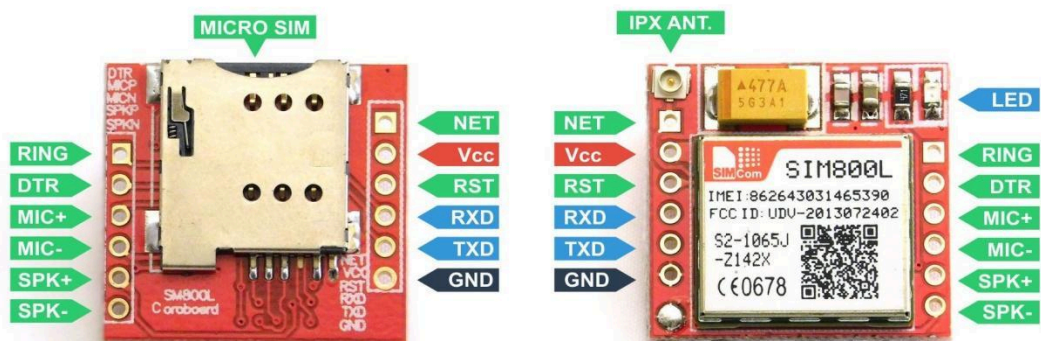
- SIM 800A Module
- SIM 808 Module
- SIM 800L Module
- SIM 800C Module

Choice of GSM Module

In this design, SIM 800L is chosen as the module to enable SMS communication between the mobile phone and main controller. It is chosen mainly because; it allows SMS communication between the system and user, it has lower cost, small footprint and it easy to develop.

SIM800L Module

SIM800L is a miniature cellular module which allows for GPRS transmission, sending and receiving SMS and making and receiving voice calls. Low cost and small footprint and quad band frequency support make this module perfect solution for any project that require long range connectivity. After connecting power module boots up, searches for cellular network and login automatically. This module has two antennas included. First is made of wire (which solders directly to NET pin on PCB) - very useful in narrow places. Second - PCB antenna - with double side tape and attached pigtail cable with IPX connector. This one has better performance and allows you to put your module inside a metal case - as long the antenna is outside.



Specification

- Supply voltage: 3.8V - 4.2V

- Recommended supply voltage: 4V
- Power consumption:
 - sleep mode < 2.0mA
 - idle mode < 7.0mA
 - GSM transmission (avg): 350 mA
 - GSM transmission (peek): 2000mA
- Module size: 25 x 23 mm
- Interface: UART (max. 2.8V) and AT commands
- SIM card socket: microSIM (bottom side)
- Supported frequencies: Quad Band (850 / 950 / 1800 /1900 MHz)
- Antenna connector: IPX
- Status signaling: LED
- Working temperature range: -40 do + 85 ° C

3.4.4. Relay

A relay is an electrically operated switch. It consists of a set of input terminals for a single or multiple control signals, and a set of operating contact terminals. The switch may have any number of contacts in multiple contact forms, such as make contacts, break contacts, or combinations thereof.

The three main types of relays are

- Electromechanical
- solid-state
- reed

Choice of Relay

In, this design, electromagnetic relay is chosen as the relay to drive the motor. It is chosen mainly because of its

- low power consumption
- low cost
- Able to withstand large inrush currents
- Able to carry high voltage, high current load
- Versatility.

Electromagnetic Relay

These relays are constructed with electrical, mechanical and magnetic components. They are made with a coil that induces a magnetic field when energized. This magnetic field attracts the armature (a movable contact) which closes or opens the contacts. When the coil is de-energized, the coil loses its magnetic field and a spring then retracts the armature to its normal position which then again, opens or closes the contacts. Electromechanical Relays are designed for AC or

DC source depending on the applications. The AC and DC relays work on the same principle as electromagnetic induction but its structure may differ in its coil construction.



Specifications of Electromagnetic Relay chosen

- Power supplied: 30A 240VAC 28VDC
- Coil Volts: 12V DC
- Number of Pins: 6

3.4.5. Real-time clock

A real-time clock (RTC) is an electronic device designed in the form of an integrated circuit to measure the passage of real-time. Real-time clocks maintain accurate time measurements within an embedded system even when the main power is off. It counts hours, minutes, seconds, months, days, and even years. These modules can be found in nearly every electronic device especially when the device requires a form of accurate timekeeping. The three-letter acronym for real-time clocks, RTC, is also shared with other computing terms like real-time computing and real-time counter. This often leads to confusion early on for beginners, especially as real-time counters are sometimes similarly used in software programming to real-time clocks. The best way to keep the terms easily separated and defined is to think of real-time clocks as physical computer chip modules usually embedded in the microcontroller unit while the other terms are used for software functions resulting from hardware performance.

Choice of Real time clock

In this design, the real time clock chosen is DS3231M.

DSP3231 RTC Module

The DS3231 is a low-cost, extremely accurate I2C real-time clock (RTC) with an integrated temperature-compensated crystal oscillator (TCXO) and crystal. The device incorporates a battery input, and maintains accurate timekeeping when main power to the device is interrupted.



DS3231 RTC Module Specifications

- Operating voltage of DS3231 MODULE: 2.3V – 5.5V
- Can operate on LOW voltages
- Consumes 500nA on battery backup
- Maximum voltage at SDA , SCL : VCC + 0.3V
- Operating temperature: -45°C to +80°C

3.4.6. Buzzer

An audio signaling device like a beeper or buzzer may be electromechanical or piezoelectric or mechanical type. The main function of this is to convert the signal from audio to sound. Generally, it is powered through DC voltage and used in timers, alarm devices, printers, alarms, computers, etc. Based on the various designs, it can generate different sounds like alarm, music, bell & siren.

C hoice of Buzzer

In this design, an electromagnetic buzzer is chosen. The designing of these types of buzzers can be done with a bare metal disc & an electromagnet. The working principle of this is similar to magnetic and electromagnetic. It generates sound throughout the disc movement & magnetism.



Specifications

- Color is black with diameter of 12mm.

- The frequency is 2kHz
- Operating Temperature ranges from -20°C to $+60^{\circ}\text{C}$
- Supply voltage is 5V
- Operating voltage ranges from 3V to 12V DC
- The sound pressure level is 85dBA or 10cm
- Current Consumption is 30mA

3.4.7. Display

Display is a device that is used to present visual information. The main of any display technology is to simplify the information sharing.

The different types of display technologies include:

- o Liquid Crystal Displays (LCD)
- o Organic Light Emitting Diodes (OLED)
- o Light Processing Technology
- o Plasma displays
- o Field Emission Displays
- o Electronic Paper

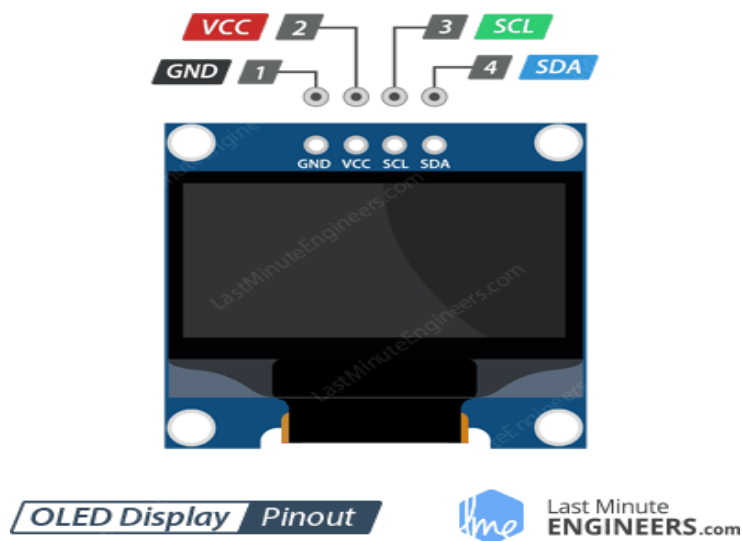
Choice of Display technology

In this design, OLED is chosen as the display technology for displaying the gate operation; who and when the gate was accessed. OLED is chosen due to its:

- o Improved image quality – better contrast
- o Low power consumption
- o Simpler design
- o Better durability – OLEDs are very durable and can operate in a broader temperature range

OLED Display

OLED (Organic Light Emitting Diodes) is a flat light emitting technology, made by placing a series of organic thin films between two conductors. When an electric current is applied, a bright light is emitted. OLEDs are used to create digital displays in devices such as television screens, computer monitors and portable systems such as smartphones and handheld game consoles.



Features

- o Number of pixels: 128 x 64
- o Color Depth: Monochrome
- o Working Voltage: 3.3 – 5V DC
- o Power: 0.06W Max
- o Interface: I2C

3.4.8. Tri-Button Manual Override

It consists of a three normally-open push button that is pressed in a preset sequence to give manual control to the user in absence of the Bluetooth Transponder and GSM Phone. It is labelled; TBMO1, TBMO2, TBMO3.

TBMO1 - Set manual override: Press and hold the button till countdown appears and stops.

TBMO2 - Enable manual override: Press and hold the button till countdown appears and stops for you to write the preset code.

TBMO3 - Reset user data: Press and hold the buton till the countdown appears and stops for the programmer to change the manual override codes.

3.5. Sliding Gate

A sliding gate is a fence that can be rolled to one side. Sliding gates provide a practical, safe, secure, and recognizable entryway to terrains. A sliding gate is a flexible fence that has wheels at the bottom. A sliding gate does not have a turning circle, so it is easy to open and close.

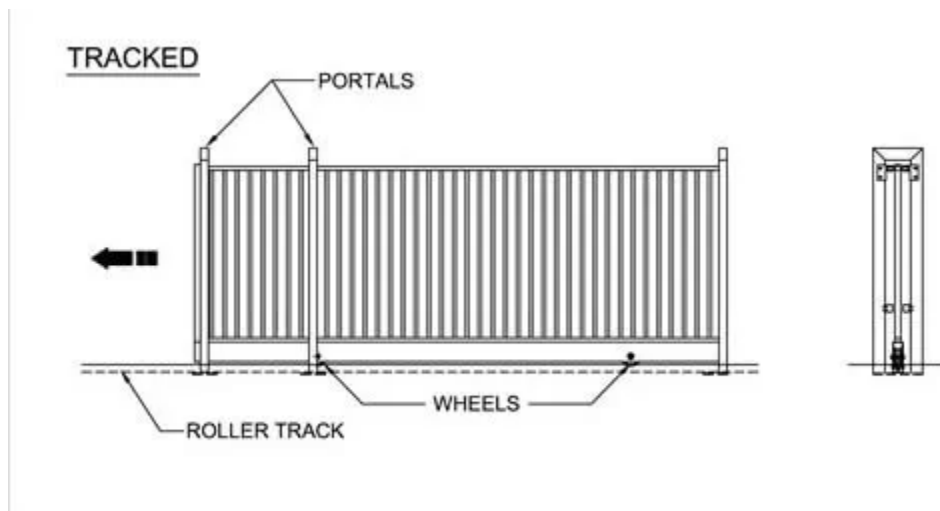
The two most common types of sliding gate:

- Tracked sliding gate
- Cantilever sliding gate

Choice of sliding gate:

In this project, the type of gate selected is the tracked sliding gate. It is chosen because:

- o It is space efficient
- o Less cost
- o Simple and robust design



Tracked sliding gate

It is a wheel-mounted gate that slides along a track. In order to operate gate automatically, there are 2 options of gate operating mechanisms.

Options

- The Gear operated or Rack and Pinion
 - Chain-driven
- o Chain-driven

This type of gate operator is ordinarily, encased in a rectangular box and is mounted to a cement pad and positioned close to what would be the tail end of the gate when the gate is closed. The motor turns a drive sprocket that opens and closes the slide gate via chain in a manner similar to raising and lowering a flag on a pole. The mechanism chain pulls and pushes the gate over the width of the driveway

Advantages

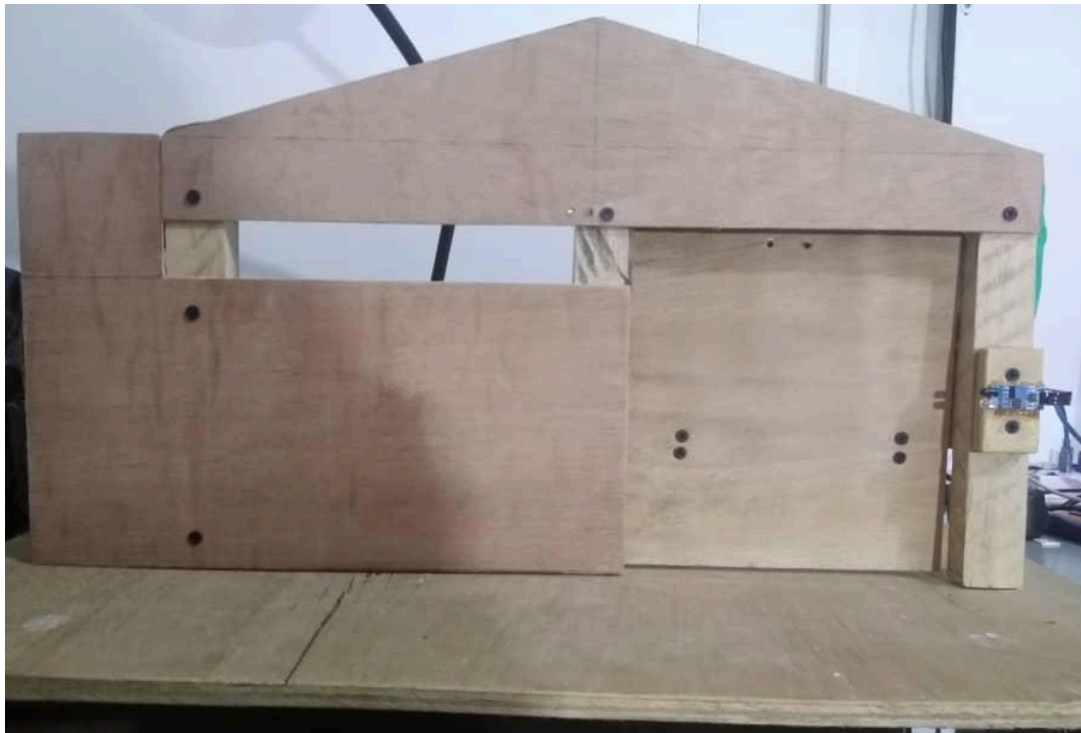
- o The chain allows flexibility in the level of the terrain.
- o Built-in battery available at extra cost

Disadvantages

- o The chain tends to stretch over time and needs periodic adjustment.
- o The chain rusts and stains the driveway.
- o The tail of the gate carries the majority of the gate weight.

Selected Sliding Gate

In this project, we select a prototype sliding gate that works on chain and sprocket mechanism.



3.6. Motor

An electric motor is an electrical machine that converts electrical energy to mechanical energy.

Options

- AC Motor
- DC Motors

Choice of Motor

In this design, the type of motor chosen is DC motor. It is selected because;

- o High starting power torque
- o Low power consumption
- o Ensures high operations (open/close) per hour as they can work continuously without overheating.

Motor Sizing

The DC Motor selected has the following specifications

- Working Voltage : 24V, 12V DC
- Type: Brush type
- Motor's Power: 14.4W, 21W
- Output torque: 5-25 kg.cm



4. METHODOLOGY & IMPLEMENTATION

4.1. INTRODUCTION

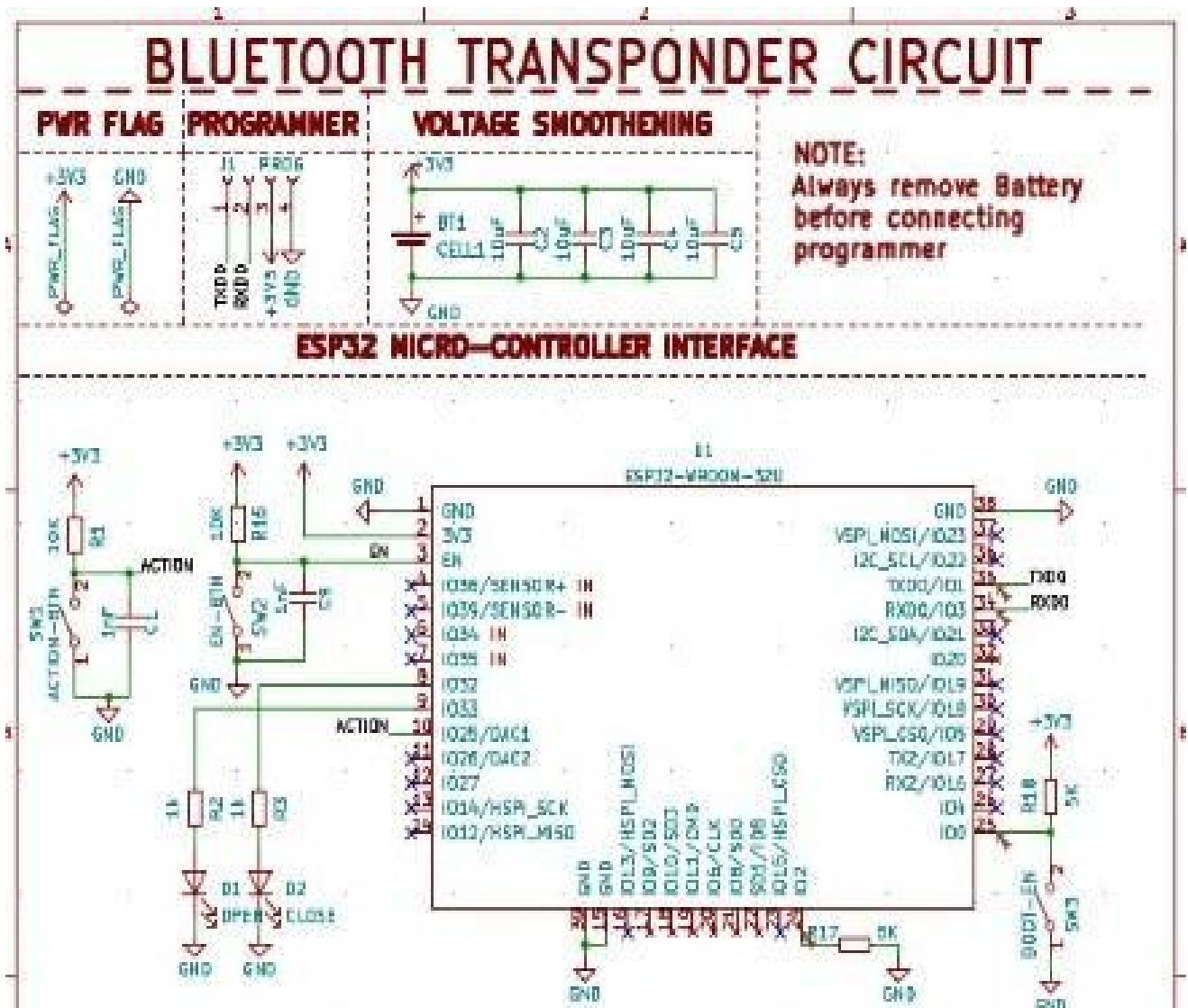
This chapter deals with relevant design methodology, the mechanism and working principles of providing motion in order to open and close the gate, it will be stated and explained stage by stage.

4.1.1. Design of the circuit diagram of the various units using Kicad software.

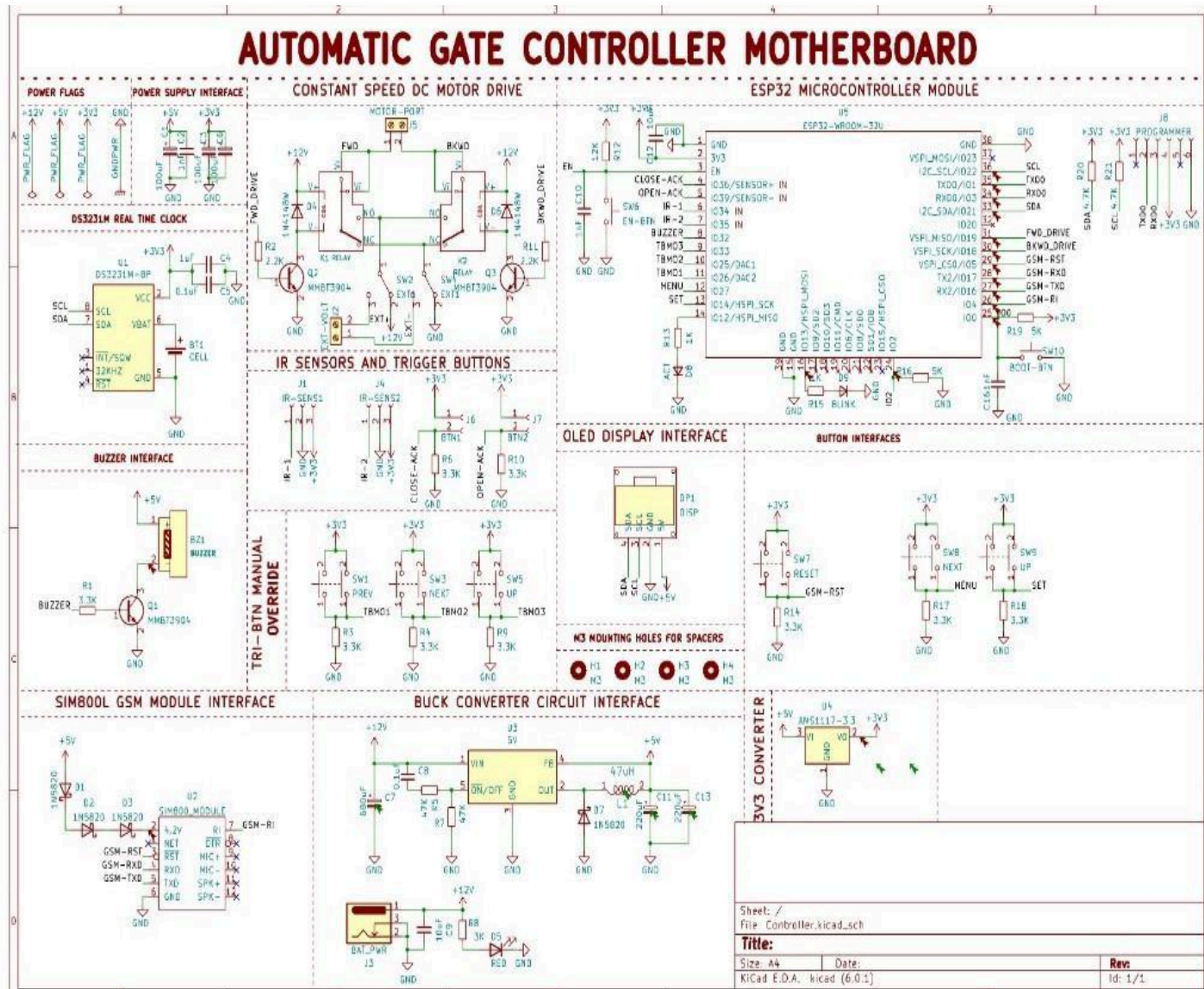
The circuit for remote automatic gate control (Bluetooth Transponder and Main Controller) is designed using Kicad 6.0 Software. The software facilitates the design and simulation of electronic the proposed design system. The figures below shows the wiring circuit designs

Comprehensive Circuit Diagram of Bluetooth Transponder

Circuit Design of the Bluetooth Transponder

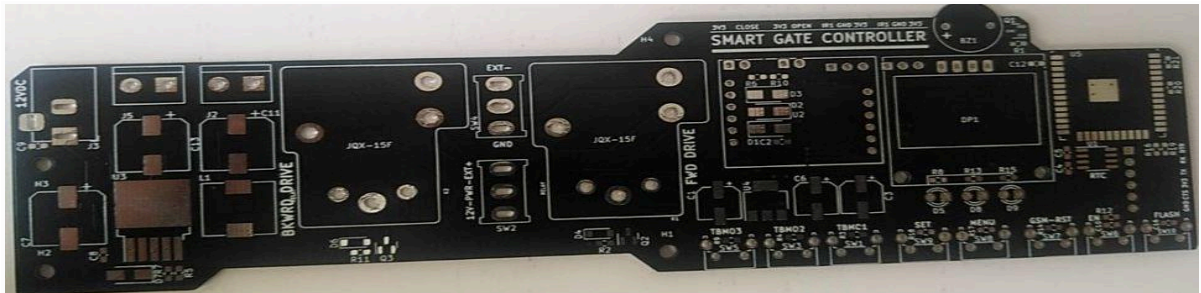
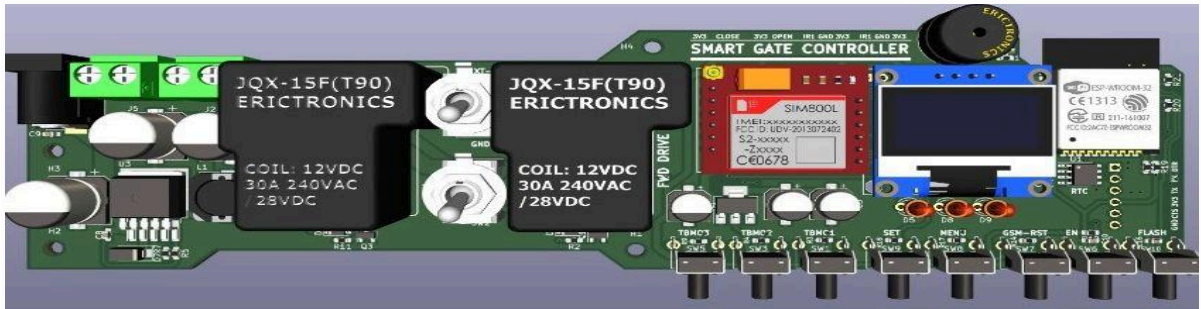
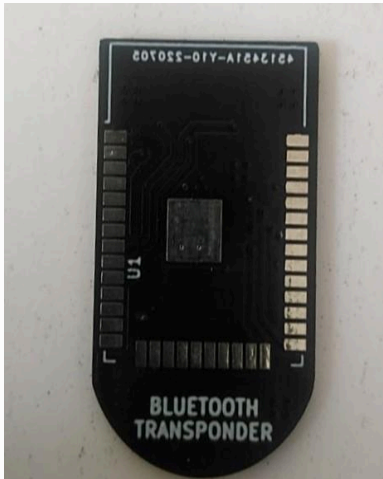
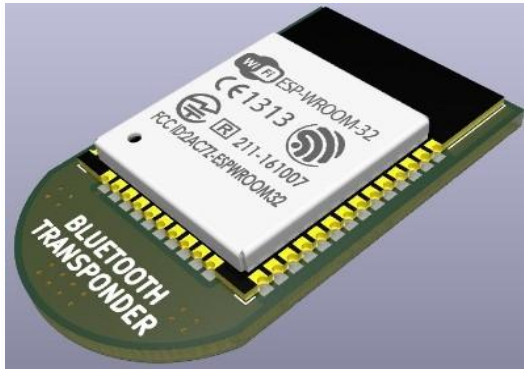
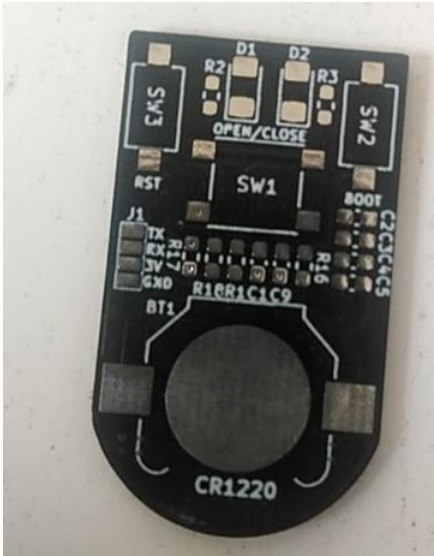
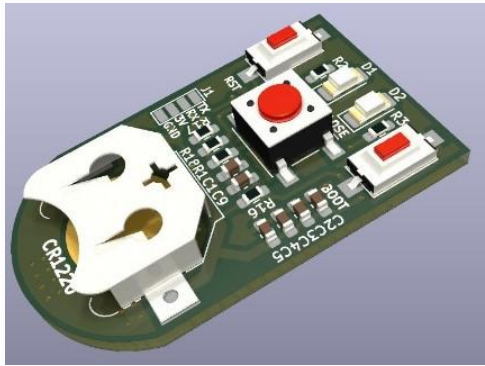


Comprehensive Circuit Diagram of Main controller



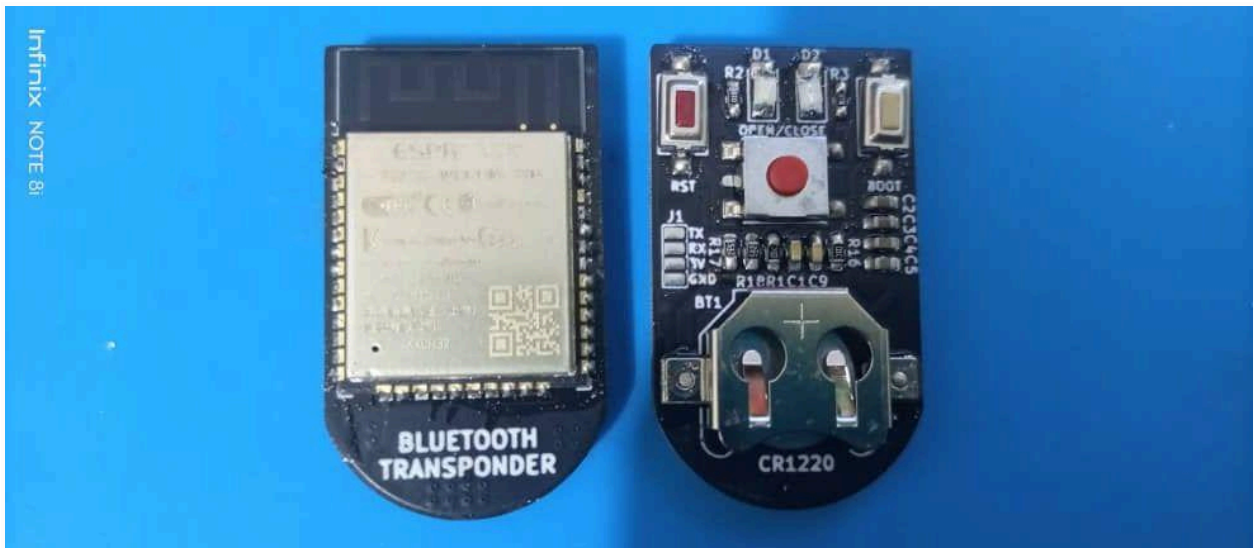
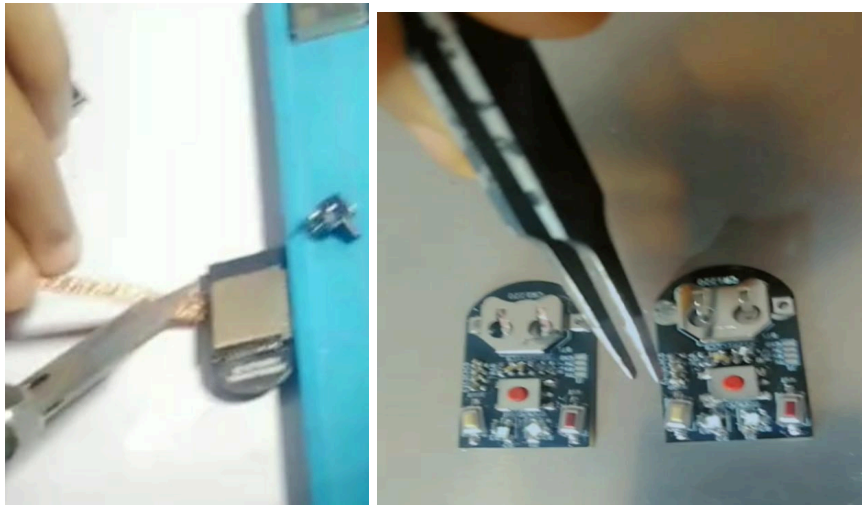
4.1.2. PCB Design and fabrication of the circuit

The 3D footprint of the of the circuit design is generated from Kicad. The 3D layout is fabricated in China and shipped to Ghana.



4.1.3. Acquisition of components and Assembly of components

The various components selected in the previous chapter is purchased. The components are soldered on the PCB board using soldering iron, lead, soldering paste and heating pad. Precaution is taken to ensure that soldering was properly done.



4.1.4. Testing the entire system: Testing the Bluetooth Transponder, Main Controller.

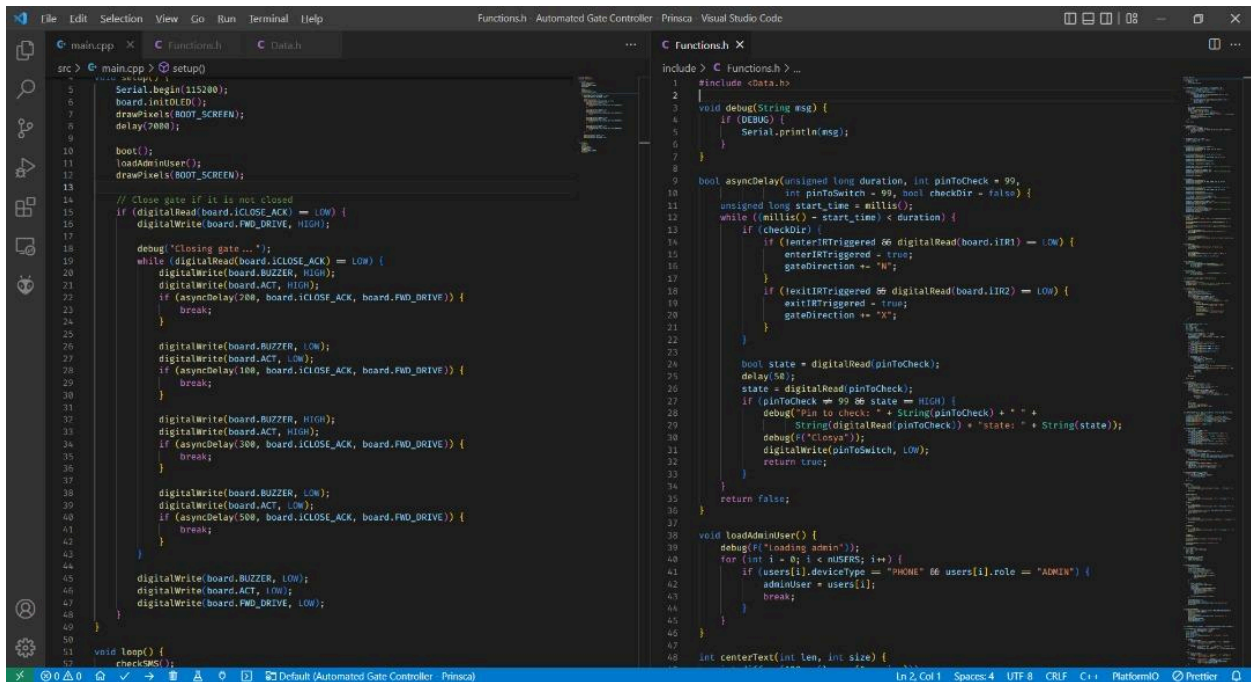
The Bluetooth controller has an operating voltage of 3.3V. The main controller has operating voltages; 12V, 5V and 3V.

A 12V DC Supply is given to the main controller through the DC barrel. The buck and linear converter installed, converts this voltage to 5V and 3V for the use of ESP32 Microcontroller, GSM Module etc.

Each stage of the the soldering process was tested using the multi-meter to ensure it is working properly before the next stage is done.

4.1.5. Software development

The computer program used to carry out the programming of the Bluetooth Transponder and Main Controller is Arduino IDE Software which was written in C-programming Language.



4.2. Mode of Operation

The remote controlled automatic gate is designed and constructed to operate on Bluetooth and SMS communication between the user, the motor controller and gate driver.

To illustrate the applicability of the theory, a prototype gate is constructed and wired up with the motor, Main Controller and Sensor(positioned carefully on two opposite sides of the gate). When the remote automatic system is powered on, one LED on the Main Controller lights red to indicate power. The system is linked with ESP32 Microcontroller and also, SIM800L GSM modem, a provider of Sim-card network services, preparing the system to accept a command using the user. When a user sends an opening command (AT+OPEN) to the SIMcard number within the SIM800L modem from a mobile device, the ESP32 Microcontroller reads the code. The microcontroller performs the function and gives command to the relays which act as constant speed drives for our motor.

The Bluetooth Transponder is programmed to auto connect with the Main Controller via Bluetooth once its powered. When the push button is pressed, the Transponder connects with the Controller. This connection is indicated on the OLED. Pressing the button after connection has been made, the gate opens. The gate closes after 1 minute once the sensor does not sense motion across the gate.

We are also providing a special feature in our system. Which is the Tri-Button Manual Override. Each of these numbers represent the numbers; 1, 2 and 3. To open the gate, TBMO2 is pressed and held till the countdown appears and completes. The preset password(123) programmed into the microcontroller is inputted, the SET button is pushed and the gate opens.

The microcontroller is programmed to recognize only UID and phone numbers included in the code. If an SMS is sent from an unknown number, access to the gate is denied.

5. RESULTS & DISCUSSION

The aim of this project work is to design and construct a remote controlled automatic gate for opening and closing the gate. The result of the processes lead to a sliding gate which can be opened by pressing the button on the Transponder, sending an SMS code through a phone or even using a manual override control.

We have checked the IR obstacle sensor. It stops the gate from closing when there's motion across the gate and also triggers the circuitry to close the gate.

The system at any point of its work, sends a timestamped notification to the administrator indicating when the door opens or closes. It also includes whether the user entered or exited the gate.

The electrical circuit is working properly and our objectives are fulfilled.

6. CONCLUSION & RECOMMENDATION

6.1. CONCLUSION

The contributions from the literature provided a wealth of knowledge and served as the theoretical foundation for this research and made it successful. Some factors such as economy, availability of components, research materials, efficiency, compatibility, portability and durability were considered during the design and fabrication stages of the project. The performance of the system after test met design specifications.

A variety of gate types can be employed with the technology. The structure used in this project has been designed and constructed to serve as a small opening to demonstrate the effectiveness of the idea.

6.2. RECOMMENDATION

The design of gates for spaces requiring high security, such as industrial warehouses, could be improved upon using the ideas illustrated by this research. In the event of a power outage, the system should also have a backup power source.

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